



Project

Summer



Umbrella



Project Name :

The name “Umbrella” is inspired by its use in the **Resident Evil universe**, where it represents a central force that unifies and controls **complex genetic and biological experimentation**. In that context, it symbolizes a system that works across DNA manipulation, mutations, and cross-species biological processes under one coordinated structure.

Problem Statement :

Biological and genomic information about animal species is fragmented across **multiple databases**, while many genomes remain **incomplete** due to DNA degradation, sequencing limitations, and assembly challenges. These missing genomic regions limit our understanding of species evolution, adaptations, and biodiversity. Traditional reconstruction methods often struggle to accurately recover large missing DNA segments.

Key question :

How can Artificial Intelligence be used to reconstruct missing DNA sequences from fragmented ancient genomes in order to provide a more complete and biologically plausible genetic blueprint of extinct species?

Who Is Affected by This Problem?

- Evolutionary Biologists
- Geneticists and Genomic Researchers
- Conservation Biologists



- Paleontologists
- Synthetic Biology and De-Extinction Researchers
- Students and Educators

Proposed Solution :

Animal BioHub is an Agentic Multi-Agent Conversational Platform that orchestrates specialized AI agents capable of answering biological questions, analyzing genomes, discovering traits, reconstructing missing DNA regions, and exploring biodiversity information.

Instead of requiring users to manually search multiple databases, the system provides a unified conversational interface where a central orchestrator delegates tasks to domain-specific agents.

Project Objectives :

- Build a centralized AI-powered platform for animal genomic information.
- Develop specialized AI agents capable of answering biological questions.
- Predict biologically plausible DNA sequences for incomplete genomes.
- Integrate multiple biological databases into a unified interface.
- Visualize protein structures and biodiversity information.
- Provide an intuitive conversational experience for researchers, students, and educators.



Specialized AI Agents ;

- **Genome Agent** : RAG + APIs +LLM + Prompting
- **Protein Structure Visualization Agent**: APIs + LLM + Prompting
- **Evolution Agent** : RAG + GraphRAG + + LLM + Prompting
- **Trait Discovery Agent** : RAG + APIs + LLM + Prompting
- **Reconstruction Agent** : Machine Learning + LLM + Prompting
- **Biodiversity Agent** : Machine Learning + LLM + Prompting
- **Literature Agent** : RAG + APIs +LLM + Prompting
- **Species Recognition Agent** : multimodal + RAG + +LLM + Prompting

Development Methodology: **Scrum**

Product Owner : Hajer AYADI

Scrum Master : Hajer AYADI & team leaders

Developers : Members & participants

Communication Channels :

- **Google Meet** : for meetings
- **Messenger** : for daily communication and announcements
- **Gmail** : for official announcements and invitations

Technologies :

- **Backend** : Python frameworks (Flask / FastApi ...)
- **Frontend** : React + Next.js
- **AI** : not limited to specific technologies
- **Databases** : PostgreSQL / ChromaDB / Qdrant / Zep MCP ...
- **Data Sources** : NCBI / UniProt / AlphaFold / GBIF / IUCN / PubMed ...



Timeline :

3 - 2 Months project (July + August + September)

Expected Deliverables :

- Functional web application
- Multi-agent backend
- Database integration pipeline
- Detailed documentation
- Final presentation
- Technical report
- GitHub repository
- Demo video

Evaluation Formula :

Final Score = $0.50 \times \text{Soutenance} + 0.15 \times \text{Technical Contribution} + 0.10 \times \text{Attendance} + 0.10 \times \text{Weekly Progress} + 0.10 \times \text{Collaboration} + 0.05 \times \text{Professionalism}$



Category	Weight	Description
Final Technical Soutenance	50 pts (50%)	Final presentation, technical understanding, demo, Q&A
Participation & Attendance	10 pts	Meetings, workshops, technical sessions
Weekly Progress & Reporting	10 pts	Regular updates, respecting milestones
Collaboration & Teamwork	10 pts	Communication, helping teammates, GitHub collaboration
Technical Contribution	15 pts	Quality and quantity of actual work
Professionalism	5 pts	Respecting deadlines, responsiveness, initiative

Project Guidelines and Policies :

- If you are unable to attend a meeting or participate in a project activity, you must inform the organizers in advance. Unexcused absences may result in removal from the Messenger group and the GitHub repository.
- If you encounter any issues, concerns, or difficulties, please communicate them promptly to me or to any member of the chapter board.
- Respecting deadlines is mandatory. All assigned tasks must be completed and submitted on time.
- Weekly progress reporting is required. Every member must regularly share updates on completed work, ongoing tasks, and overall progress.
- A team leader may be assigned to each group of 3–4 members to facilitate coordination and communication.
- Attendance at workshops is mandatory, as they are an essential part of the project.
- A scoring system will be implemented throughout the project. Evaluation criteria will include attendance, task completion, contributions, and initiative. An evaluation rubric will be shared with all participants.
- At the end of the project, a technical defense (soutenance) will be organized before a jury. Each member will be required to present and defend their work.
- **Professionalism is expected at all times, including communication, collaboration, commitment, and work quality.**

